

JUSTIN BOEHNEN

Senior Software Engineer

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Platform architect and technical lead who owns the full lifecycle of distributed systems serving 1,200+ users across web and mobile clients. Designed and executed a zero-downtime monolith-to-microservices migration, established the AI integration strategy for production APIs, and set the engineering standards for the engineering team. Independently architected and shipped BrainGap, an IRT-based adaptive assessment platform, from concept through public beta.

EXPERIENCE

Cloud Development Lead · Senior Software Engineer

Alpine Land Information Services | 2022 – Present | Remote | Engineering Team

- Inherited a monolithic API where deployments regularly broke unrelated services. Decomposed into microservices using dual-write sync and incremental cutover, serving 500k+ daily requests for 1,200+ users across mobile, desktop, and web clients. Platform supports forestry data collection for the largest private landowner in the United States. Zero downtime, zero lost accounts, 40% faster release cycles.
- Platform had zero AI capabilities. Singlehandedly integrated AI across the entire data collection and analysis pipeline: MCP with OAuth for agent workflows, LLM-powered analysis at multiple stages, and an ML model for tree height prediction that saves ~20 minutes per location across thousands of sites per season.
- Older form apps stored binaries in the database, broke submissions on template edits, and lacked spatial types. Redesigned with PostGIS queries, blob storage offloading, and template versioning. Crews now collect on one version while offices publish updates without data loss.
- Production issues were found by users before engineering knew. Established the observability practices across all services with structured logging, alerting, and dashboards. Cut mean time to recovery ~50% and shifted from reactive firefighting to proactive detection.
- Owned the full frontend surface across Blazor and JavaScript web applications and WPF/MVVM desktop clients wired into my backend APIs. Modernized legacy desktop applications by replacing direct database connections with API integrations, enabling cloud-first architecture without disrupting existing user workflows.

Cardiac Monitor Technician

Sky Lakes Medical Center | 2021 – 2022 | Klamath Falls, OR

Measured and diagnosed cardiac rhythms and vitals across a high-acuity patient load, working directly with care teams to identify critical changes and escalate appropriately. Gained firsthand understanding of clinical workflows, EHR systems (Epic), and how software design decisions directly affect patient outcomes.

INDEPENDENT ENGINEERING

[BrainGap.io](https://braingap.io) · Adaptive Assessment Platform · Public Beta

Sole Architect | .NET Aspire, Blazor, PostgreSQL (pgvector), Redis, Anthropic APIs

API-first platform that decomposes any subject into assessable concepts and measures proficiency using Item Response Theory with Maximum Fisher Information. Self-improving feedback loop where concepts split or merge from usage signals, with tiered model architecture for cost efficiency.

[ClipMap.io](https://clipmap.io) · Geospatial Data Pipeline & Web Application

Sole Architect | React, TypeScript, Cloudflare R2, QGIS Python

Eliminated backend infrastructure by migrating all geospatial processing to browser-side execution. Pre-computed 70,000+ multi-LOD tiles from 95GB OSM data served via Cloudflare R2 CDN. Zero hosting costs, unlimited concurrent exports, and infinite horizontal scale through client-side polygon operations and SVG rendering.

TECHNICAL SKILLS

Languages

C#, .NET, ASP.NET Core, WPF, Blazor, TypeScript, JavaScript, React, Node.js, Python

Cloud

Azure (App Services, Functions, Service Bus, DevOps, Insights, Storage), Docker

Data

PostgreSQL (pgvector), PostGIS, Redis, MongoDB, Azure SQL, EF Core, GIS & Spatial Data

AI

MCP (Model Context Protocol), LLM Integration, Prompt Engineering, ML Development

Architecture

Microservices, Event-Driven, Distributed Systems, MVVM, Scalable API Design

DevOps

CI/CD (GitHub Actions, Azure DevOps), Automated Testing, Monitoring, Observability